Business Case: FSx for Windows Cost Optimization via Decommissioning and Right-Sizing

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Executive Summary

Amazon FSx for Windows File Server provides highly available, managed SMB storage for Windows Kaizen legacy workloads.
However, the lack of support for storage downscaling results in escalating costs as unused or stale data accumulates.

Between **Dec 2024 and May 2025**, FSx incurred **$72,395.03** in storage costs, with a **54% increase** over six months (from $10,137.44 in Feb to $15,719.94 in May).

This proposal recommends decommissioning the over-provisioned FSx file system and deploying a **right-sized, cost-optimized replacement**, with **cold data archived to S3**. Combined with **tiering, deduplication, and lifecycle automation**, this strategy can reduce monthly costs by up to **70%**, while improving long-term sustainability.

Historical FSx Cost Summary

Month	FSx Cost (USD)
Dec 2024	$10,222.54
Jan 2025	$10,240.95
Feb 2025	$10,137.44
Mar 2025	$11,430.78
Apr 2025	$14,643.39
May 2025	$15,719.94
Total	**$72,395.03**

Extrapolated Future Costs (Status Quo)

Assuming continued 8–10% monthly growth due to data and backup accumulation:

Month	Projected FSx Cost
Jun 2025	$17,134.00
Jul 2025	$18,510.72
Aug 2025	$19,992.16
Sep 2025	$21,591.53
Oct 2025	$23,318.85

Month	Projected FSx Cost
Nov 2025	$25,184.41
6-Month Total	**$125,732.67**

Note: Without intervention, annual FSx costs could exceed **$198,000**.

Proposed Solution: FSx Decommission & Right-Sizing

Key Goals

- Reduce monthly FSx costs by up to 70%
- Align provisioned storage to actual usage
- Archive stale data to S3/Glacier for long-term retention
- Establish an automated data lifecycle process

Target Architecture Sizing

Component	Current FSx	Proposed FSx	Change
Provisioned Storage	~15 TB	6–7 TB	-50–60%
Active Data Retained	6 TB	6 TB	Same
Cold Data Archived	N/A	9 TB → S3	+S3 (~$40/mo)
Throughput Capacity	128 MBps	32 MBps	-75%
Monthly FSx Cost	~$15,700	~$4,800	-69%

Implementation Approach

Phase	Description
1. Audit & Analysis	Inventory FSx with PowerShell & TreeSize; classify data by last access date (>60 days = cold)
2. Archive Cold Data to S3	Move stale files to `Z:\archive` with PowerShell; use AWS DataSync to S3; apply S3 lifecycle (Glacier after 30 days, delete after 1 year)
3. Provision New FSx	Use SSD + HDD tiering; enable deduplication; select optimal throughput (e.g., 32 MBps)
4. Migrate Active Data	Use Robocopy or AWS DataSync for file migration; reapply NTFS permissions and share settings
5. Cutover & Decommission	Update mount points or DFS targets; verify integrity; delete old FSx volume and snapshots

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ROI & Cost Savings

Category	Current (Annual)	Proposed (Annual)	Estimated Savings
FSx Storage	$180,000+	~$57,600	~$122,400
S3 (archive data)	~$0	~$500–1,000	Minimal cost
Total Savings	-	-	**~$120K+**

ROI is expected within **60 days** with **sustained savings year-over-year**.

Strategic Benefits

Benefit	Description
Immediate cost reduction	Reduces FSx spend by 60–70%
Sustainable architecture	Aligns resources to real usage and growth trajectory
Improved data hygiene	Removes stale and redundant data
Backup efficiency	Smaller, faster, cheaper backups
Governance & compliance	Lifecycle policies improve retention management

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Risk & Mitigation

Risk	Mitigation
Downtime during migration	Schedule out-of-hours; use Robocopy with retry/mirror
Data integrity loss	Use checksums, logging, and validation scripts
Permission mismatch	Export and reapply NTFS ACLs and share settings

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Proposed Timeline

Week	Activity
1	Audit FSx, gather usage data, align with stakeholders
2	Archive and sync cold data to S3
3	Provision new FSx and migrate active data
4	Cutover, verify, and decommission old FSx

KPIs for Success

KPI	Target
% Storage Reduction	≥ 50%

KPI	Target
Monthly FSx Savings	≥ $10,000
Archived Data Volume	≥ 8 TB to S3
Cutover Downtime	< 1 hour
Stakeholder Approval	Full sign-off